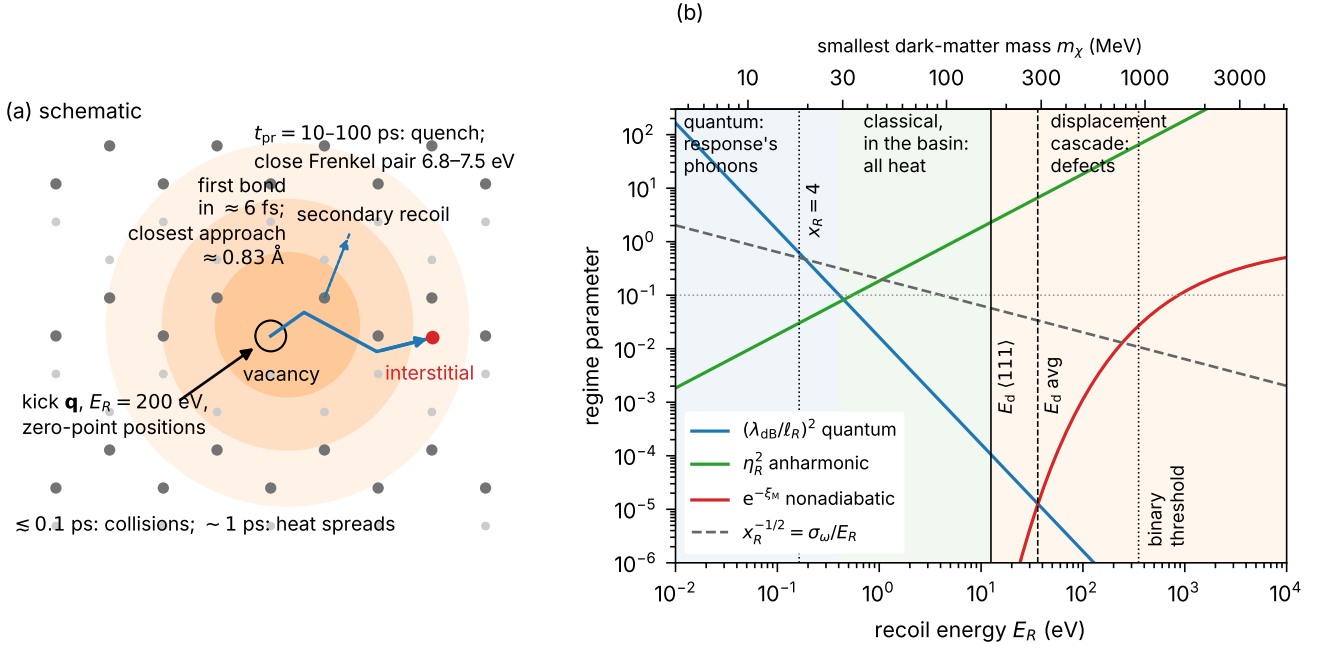
(a) schematic
t_{pr} = 10–100 ps: quench;
close Frenkel pair 6.8–7.5 eV
first bond
in ≈ 6 fs;
secondary recoil
closest approach
≈ 0.83 Å
vacancy
interstitial
kick **q**, E_R = 200 eV,
zero-point positions
≲ 0.1 ps: collisions; ~ 1 ps: heat spreads
(b)
smallest dark-matter mass m_χ (MeV)
10
30
100
300
1000
3000
quantum:
response's
phonons
classical,
in the basin:
all heat
displacement
cascade:
defects
$x_R = 4$
E_d (111)
E_d avg
binary
threshold
regime parameter
$(\lambda_{dB}/\ell_R)^2$ quantum
η_R^2 anharmonic
$e^{-\xi_M}$ nonadiabatic
$x_R^{-1/2} = \sigma_\omega/E_R$
recoil energy E_R (eV)